

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Describe the structure of a cell membrane and explain how it controls what enters the cell.

[6 marks]

2

Explain the difference between diffusion, osmosis, and active transport, giving one example of each.

[6 marks]

3

Describe the role of enzymes and explain why activity falls sharply above an optimum temperature.

[5 marks]

4

Explain how the structure of an alveolus is adapted for gas exchange.

[5 marks]

5

Outline the process of photosynthesis and state the word equation.

[5 marks]

6

Describe how a reflex arc produces a rapid response, naming the neurones involved.

[6 marks]

7

Explain how natural selection can lead to antibiotic resistance in bacteria.

[6 marks]

8

Describe an experiment to investigate the effect of light intensity on the rate of photosynthesis.

[6 marks]
